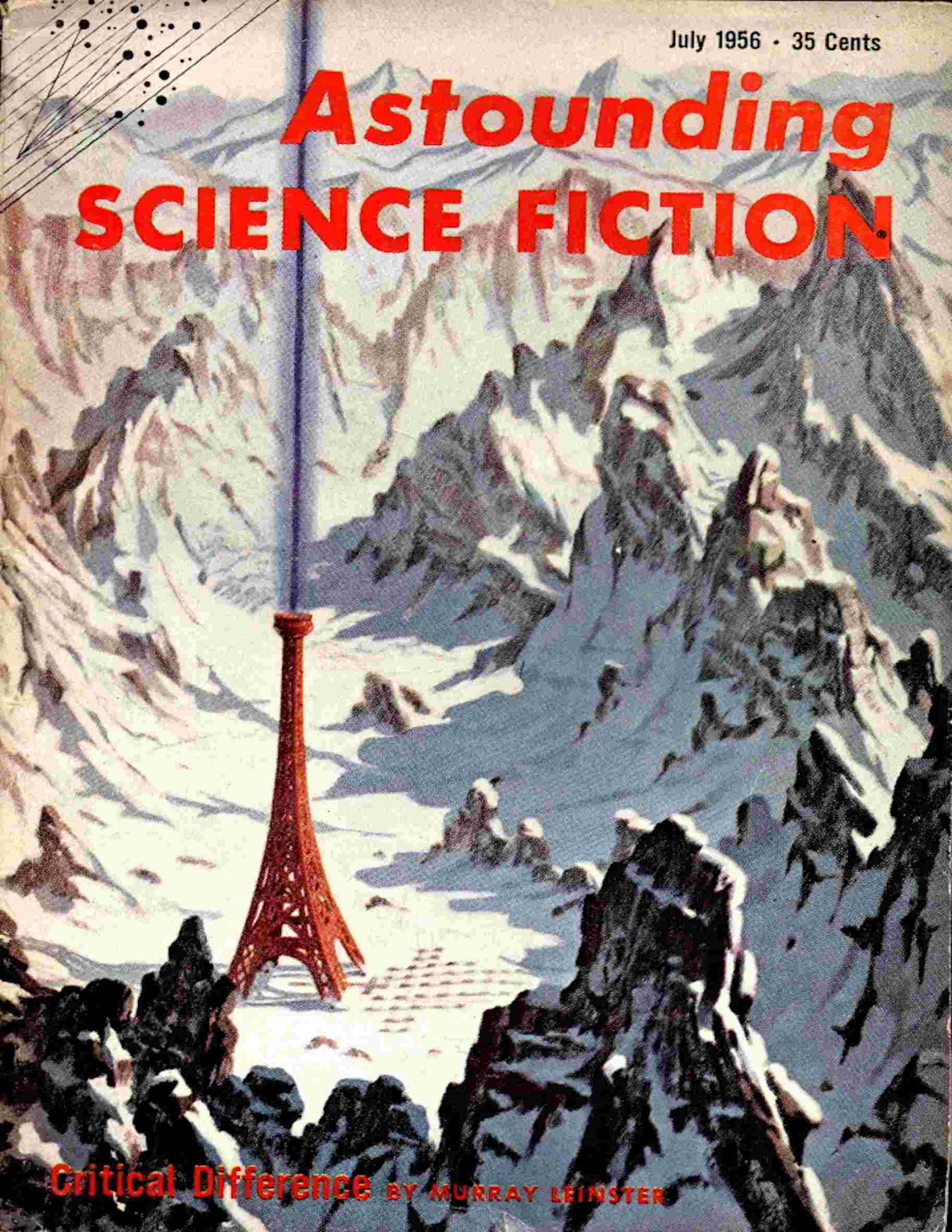


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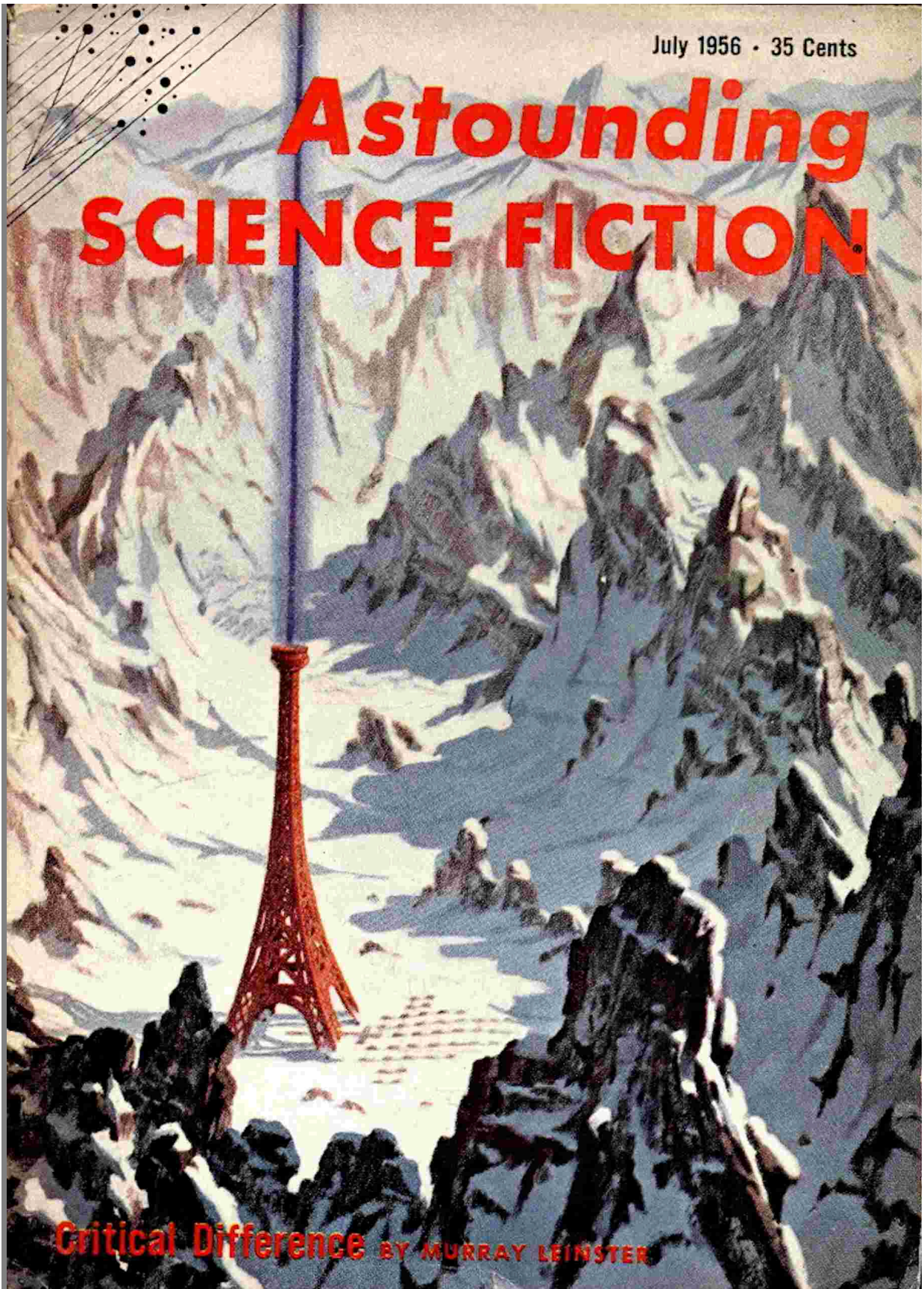
Astounding SCIENCE FICTION



Critical Difference BY MURRAY LEINSTER

July 1956 • 35 Cents

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*** START OF THE PROJECT GUTENBERG EBOOK CRITICAL
DIFFERENCE ***

CRITICAL DIFFERENCE

BY MURRAY LEINSTER

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I

Massy waked that morning when the only partly-opened port of his sleeping-cabin closed of itself and the room-warmer began to whirl. He found himself burrowed deep under his covering, and when he got his head out of it the already-bright room was bitterly cold and his breath made a fog about him.

He thought uneasily, *It's colder than yesterday!* But a Colonial Survey officer is not supposed to let himself seem disturbed, in public, and the only way to follow that rule is to follow it in private, too. So Massy composed his features, while gloom filled him. When one has just received senior service rating and is on one's very first independent survey of a new colonial installation, the unexpected can be appalling. The unexpected was definitely here, on Lani III.

He'd been a Survey Candidate on Khali II and Taret and Arepo I, all of which were tropical, and a Junior Officer on Menes III and Thotmes—one a semiarid planet and the other temperate-volcanic—and he'd done an assistant job on Saril's solitary world, which was nine-tenths water. But this first independent survey on his own was another matter. Everything was wholly unfamiliar. An ice planet with a minus point one habitability rating was upsetting in its peculiarities. He knew what the books said about glacial-world conditions, but that was all.

The denseness of the fog his breath made seemed to grow less as the room-warmer whirled and whirled. When by the thinness of the mist he guessed the temperature to be not much under freezing, he climbed out of his bunk and went to the port to look out. His cabin, of course, was in one of the drone-hulls that had brought the colony's equipment to Lani III. The other emptied hulls were precisely ranged in order outside. They were duly connected by tubular galleries, and very painstakingly leveled. They gave an impression of impassioned tidiness among the upheaved, ice-coated mountains all about.



He gazed down the long valley in which the colony lay. There were monstrous slanting peaks on either side. They partly framed the morning sun. Their sides were ice. The flanks of every mountain in view were ice. The sky was pale. The sun had four sun-dogs placed geometrically about it. It shone coldly upon this far-out world. Normal post-midnight temperatures in this valley ranged around ten below zero—and this was technically summer. But it was colder than ten below zero now. At noon there were normally tiny trickling rills of surface-thaw running down the sunlit sides of the mountains—but they froze again at night and the frost replaced itself after sunset. And this was a sheltered valley—warmer than most of the planet's surface. The sun had its sun-dogs every day, on rising. There were nights when the brighter planets had star-pups, too.

The phone-plate lighted and dimmed and lighted and dimmed. They did themselves well on Lani III—but the parent world was in this same solar system. That was rare. Massy stood before the plate and it cleared. Herndon's face peered unhappily out of it. He was even younger than Massy, and inclined to lean heavily on the supposedly vast experience of a Senior Officer of the Colonial Survey.

"Well?" said Massy—and suddenly felt very undignified in his sleeping-garments.

"We're picking up a beam from home," said Herndon anxiously, "but we can't make it out."

Because the third planet of the sun Lani was being colonized from the second, inhabited world, communication with the colony's base was possible. A tight beam could span a distance which was only light-minutes across at conjunction, and not much over a light-hour at opposition—as now. But the beam communication had been broken for the past few weeks, and shouldn't be possible again for some weeks more. The sun lay between. One couldn't expect normal sound-and-picture transmission until the parent planet had moved past the scrambler-fields of Lani. But something had come through. It would be reasonable for it to be pretty well hashed when it arrived.

"They aren't sending words or pictures," said Herndon uneasily. "The beam is wabbly and we don't know what to make of it. It's a signal, all right, and on the regular frequency. But there are all sorts of stray noises, and still in the midst of it there's some sort of signal we can't make out. It's like a whine, only it stutters. It's a broken-up sound of one pitch."

Massy rubbed his chin reflectively. He remembered a course in information theory just before he'd graduated from the Service Academy. Signals made by pulses, and pitch-changes and frequency-variations. Information was what couldn't be predicted without information. And he remembered with gratitude a seminar on the history of communication, just before he'd gone out on his first field job as a Survey Candidate.

"Hm-m-m," he said with a trace of self-consciousness. "Those noises—the stuttering ones. Would they be, on the whole, of no more than two different durations? Like—hm-m-m—*Bzz bzz bzzzzzz bzz?*"

He felt that he lost dignity by making such ribald sounds. But Herndon's face brightened.

"That's it!" he said relievedly. "That's it! Only they're high-pitched like—" His voice went falsetto. "*Bzz bzz bzz bzzzzzz bzz bzz!*"

It occurred to Massy that they sounded like two idiots. He said with dignity:

"Record everything you get, and I'll try to decode it." He added: "Before there was voice communication there were signals by light and sounds in groups of long and short units. They came in groups, to stand for letters, and things were spelled out. Of course there were larger groups which were words. Very crude system, but it worked when there was great interference, as in the early days. If there's some emergency, your home world might try to get through the sun's scrambler-field that way."

"Undoubtedly!" said Herndon, with even greater relief. "No question, that's it!"

He regarded Massy with great respect as he clicked off. His image faded. The plate was clear.

He thinks I'm wonderful, thought Massy wryly. Because I'm Colonial Survey. But all I know is what's been taught me. It's bound to show up sooner or later. Damn!

He dressed. From time to time he looked out the port again. The intolerable cold of Lani III had intensified, lately. There was some idea that sunspots were somehow the cause. He couldn't make out sunspots with the naked eye, but the sun did look pale, with its accompanying sun-dogs. Massy was annoyed by them. They were the result of microscopic ice-crystals suspended in the air. There was no dust on this planet, but there was plenty of ice! It was in the air and on the ground and even under it. To be sure, the drills for the foundation of the great landing-grid had brought up cores of frozen humus along with frozen clay, so there must have been a time when this world had known clouds and seas and vegetation. But it was millions, maybe hundreds of millions of years ago. Right now, though, it was only warm enough to have an atmosphere and very slight and partial thawings in direct sunlight, in sheltered spots, at midday. It couldn't support life, because life is always dependent on other life, and there is a temperature below which a natural ecological system can't maintain itself. The past few weeks, the climate had been such that even human-supplied life looked dubious.

Massy slipped on his Colonial Survey uniform with its palm-tree insignia. Nothing could be much more inappropriate than palm-tree symbols on a

planet with sixty feet of permafrost. Massy reflected wryly, *The construction gang calls it a blast, instead of a tree, because we blow up when they try to dodge specifications. But specifications have to be met! You can't bet the lives of a colony or even a ship's crew on half-built facilities!*

He marched down the corridor from his sleeping room, with the dignity he painstakingly tried to maintain for the sake of the Colonial Survey. It was a pretty lonely business, being dignified all the time. If Herndon didn't look so respectful, it would have been pleasant to be more friendly. But Herndon revered him. Even his sister Riki—

But Massy put her firmly out of his mind. He was on Lani III to check and approve the colony installations. There was the giant landing-grid for spaceships, which took power from the ionosphere to bring heavily loaded space-vessels gently to the ground, and in between times took power from the same source to supply the colony's needs. It also lifted visiting spacecraft the necessary five planetary diameters out when they took off again. There was power-storage in the remote event of disaster to that giant device. There was a food-reserve and the necessary resources for its indefinite stretching in case of need. That usually meant hydroponic installations. There was a reason for the colony, which would make it self-supporting—here a mine. All these things had had to be finished and operable and inspected by a duly qualified Colonial Survey officer before the colony could be licensed for unlimited use. It was all very normal and official, but Massy was the newest Senior Survey Officer on the list, and this was the first of his independent operations. He felt inadequate, sometimes.

He passed through the vestibule between this drone-hull and the next. He went directly to Herndon's office. Herndon, like himself, was newly endowed with authority. He was actually a mining-and-minerals man and a youthful prodigy in that field, but when the director of the colony was taken ill while a supply ship was aground, he went back to the home planet and command devolved on Herndon. *I wonder*, thought Massy, *if he feels as shaky as I do?*

When he entered the office, Herndon sat listening to a literal hash of noises coming out of a speaker on his desk. The cryptic signal had been relayed to

him, and a recorder stored it as it came. There were cracklings and squeals and moaning sounds, and sputters and rumbles and growls. But behind the façade of confusion there was a tiny, interrupted, high-pitched noise. It was a monotone whining not to be confused with the random sounds accompanying it. Sometimes it faded almost to inaudibility, and sometimes it was sharp and clear. But it was a distinctive sound in itself, and it was made up of short whines and longer ones of two durations only.

"I've put Riki at making a transcription of what we've got," said Herndon with relief as he saw Massy. "She'll make short marks for the short sounds, and long ones for the long. I've told her to try to separate the groups. We've got a full half hour of it, already."

Massy made an inspired guess.

"I would expect it to be the same message repeated over and over," he said. He added. "And I think it would be decoded by guessing at the letters in two-letter and three-letter words, as clues to longer ones. That's quicker than statistical analysis of frequency."

Herndon instantly pressed buttons under his phone-plate. He relayed the information to Riki, his sister, as if it were gospel. Massy remembered guiltily that it wasn't gospel. It was simply a trick recalled from his boyhood, when he was passionately interested in secret languages. His interest had faded when he realized he had no secrets to record or transmit.

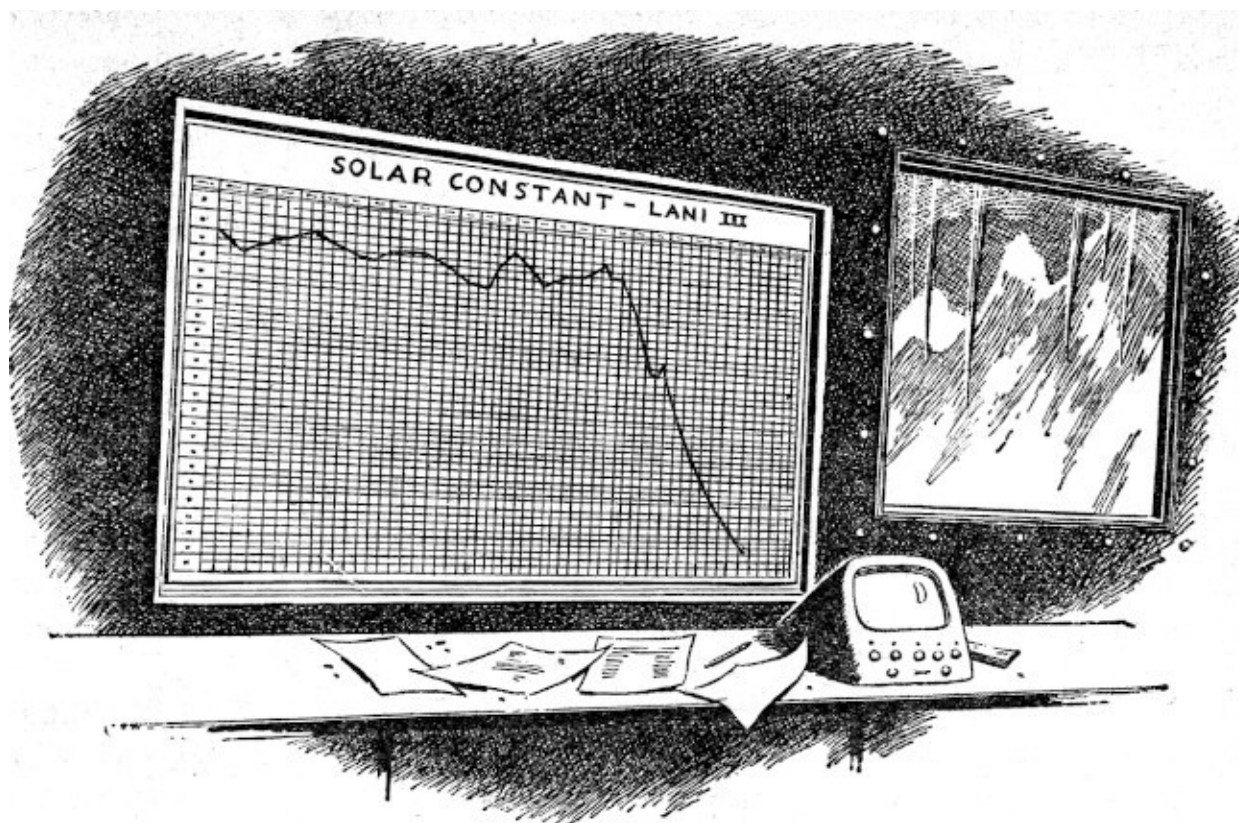
Herndon turned from the phone-plate.

"Riki says she's already learned to recognize some groups," he reported, "but thanks for the advice. Now what?"

Massy sat down. He'd have liked some coffee, but he was being treated with such respect that the role of demigod was almost forced on him.

"It seems to me," he observed, "that the increased cold out here might not be local. Sunspots—"

Herndon jittered visibly. He silently handed over a sheet of paper with observation-figures on top and a graph below them which related the observations to each other. They were the daily, at-first-routine, measurements of the solar constant from Lani III. The graph-line almost ran off the paper at the bottom.



"To look at this," he admitted, "you'd think the sun was going out. Of course it can't be," he added hastily. "Not possibly! But there is an extraordinary number of sunspots. Maybe they'll clear. But meanwhile the amount of heat reaching us is dropping. As far as I know there's no parallel to it. Night temperatures are thirty degrees lower than they should be. Not only here, either, but at all the robot weather stations that have been spotted around the planet. They average forty below zero minimum, instead of ten. And—there is that terrific lot of sunspots...."

He looked hopefully at Massy. Massy frowned. Sunspots are things about which nothing can be done. Yet the habitability of a borderline planet, anyhow, can very well depend on them. An infinitesimal change in sun heat can make a serious change in any planet's temperature. In the books, the ancient mother planet Earth was said to have entered glacial periods through

a drop of only three degrees in the planet-wide temperature, and to have been tropic almost to its poles from a rise of only six. It had been guessed that glacial periods in the planet where humanity began had been caused by coincidences of sunspot maxima.

This planet was already glacial to its equator. There was a genuinely abnormal number of sunspots on Lani, its sun. Sunspots could account for worsening conditions here, perhaps. *That message from the inner planet could be bad*, thought Massy, *if the solar constant drops and stays down a while*. But aloud he said:

"There couldn't be a really significant permanent change. Not quickly, anyhow. Lani's a Sol-type star, and they aren't variables, though of course any dynamic system like a sun will have cyclic modifications of one sort or another. But they usually cancel out."

He sounded encouraging, even to himself. But there was a stirring behind him. Riki Herndon had come silently into her brother's office. She looked pale. She put papers down on her brother's desk.

"But," she said evenly, "while cycles sometimes cancel, sometimes they enhance each other. They heterodyne. That's what's happening."

Massy scrambled to his feet, flushing. Herndon said sharply:

"What? Where'd you get that stuff, Riki?"

She nodded at the sheaf of papers she'd just laid down.

"That's the news from home." She nodded again, to Massy. "You were right. It was the same message, repeated over and over. And I decoded it like children decode each other's secret messages. I did that to Ken once. He was twelve, and I decoded his diary, and I remember how angry he was that I'd found out he didn't have any secrets."

She tried to smile. But Herndon wasn't listening. He read swiftly. Massy saw that the under sheets were rows of dots and dashes, painstakingly transcribed and then decoded. There were letters under each group of marks.

Herndon was very white when he'd finished. He handed the sheet to Massy. Riki's handwriting was precise and clear. Massy read:

"FOR YOUR INFORMATION THE SOLAR CONSTANT IS DROPPING RAPIDLY DUE TO COINCIDENCE OF CYCLIC VARIATIONS IN SUNSPOT ACTIVITY WITH PREVIOUS UNOBSERVED LONG CYCLES APPARENTLY INCREASING THE EFFECT MAXIMUM IS NOT YET REACHED AND IT IS EXPECTED THAT THIS PLANET WILL BECOME UNINHABITABLE FOR A TIME ALREADY KILLING FROSTS HAVE DESTROYED CROPS IN SUMMER HEMISPHERE IT IS IMPROBABLE THAT MORE THAN A SMALL PART OF THE POPULATION CAN BE SHELTERED AND WARMED THROUGH DEVELOPING GLACIAL CONDITIONS WHICH WILL REACH TO EQUATOR IN TWO HUNDRED DAYS THE COLD CONDITIONS ARE COMPUTED TO LAST TWO THOUSAND DAYS BEFORE NORMAL SOLAR CONSTANT RECURS THIS INFORMATION IS SENT YOU TO ADVISE IMMEDIATE DEVELOPMENT OF HYDROPONIC FOOD SUPPLY AND OTHER PRECAUTIONS MESSAGE ENDS FOR YOUR INFORMATION THE SOLAR CONSTANT IS DROPPING RAPIDLY DUE TO COINCIDENCE OF CYCLIC—"

Massy looked up. Herndon's face was ghastly. Massy said in some grimness:

"Kent IV's the nearest world your planet could hope to get help from. A mail liner will make it in two months. Kent IV might be able to send three ships—to get here in two months more. That's no good!"

He felt sick. Human-inhabited planets are far apart. The average distance of stars of all types—there is on an average between four and five light-years of distance between suns. They are two months' spaceship journey apart. And not all stars are sol-type or have inhabited planets. Colonized worlds are like isolated islands in an unimaginably vast ocean, and the ships that ply between them at thirty light-speeds seem merely to creep. In ancient days on the mother planet Earth, men sailed for months between ports, in their clumsy sailing ships. There was no way to send messages faster than they could travel. Nowadays there was little improvement. News of the Lani disaster could not be transmitted. It had to be carried, as between stars, and carriage was slow and response to news of disaster was no faster.

The inner planet, Lani II, had twenty millions of inhabitants, as against the three hundred people in the colony on Lani III. The outer planet was already frozen, but there would be glaciation on the inner world in two hundred days. Glaciation and human life are mutually exclusive. Human beings can survive only so long as food and power hold out, and shelter against really bitter cold cannot be improvised for twenty million people! And, of course, there could be no outside help on any adequate scale. News of the need for it would travel too slowly. One other world might hear in two months, and send what aid it could in four. But the next would not hear for four months, and could not send help in less than eight. It would take five Earth-years to get a thousand ships to Lani II—and a thousand ships could not rescue more than one per cent of the population. But in five years there would not be nearly so many people left alive.

Herndon licked his lips. There were three hundred people in the already-frozen colony. They had food and power and shelter. They had been considered splendidly daring to risk the conditions here. But all their home world would presently be like this. And there was no possibility of equipping everybody there as the colonists were equipped.

"Our people," said Riki in a thin voice, "all of them.... Mother and father and—the others. Our cousins. All our friends. Home is going to be like ... like that!"

She jerked her head toward a port which let in the frigid colony-world's white daylight. Her face worked.

Massy was aware of an extreme unhappiness on her account. For himself, of course, the tragedy was less. He had no family. He had very few friends. But he could see something that had not occurred to them as yet.

"Of course," he said, "it's not only their trouble. If the solar constant is really dropping like that ... why things out here will be pretty bad too. A lot worse than they are now. We'll have to get to work to save ourselves!"

Riki did not look at him. Herndon bit his lips. It was plain that their own fate did not concern them immediately. But when one's home world is doomed, one's personal safety seems a very trivial matter.

There was silence save for the crackling, tumultuous noises that came out of the speaker on Herndon's desk. In the midst of that confused sound there was a wavering, whining, high-pitched note which swelled and faded and grew distinct again.

"We," said Massy without confidence, "are right now in the conditions they'll face a good long time from now."

Herndon said dully:

"But we couldn't live here without supplies from home. Or even without the equipment we brought. But they can't get supplies from anywhere, and they can't make such equipment for everybody! They'll die!" He swallowed, and there was a clicking noise in his throat. "They ... they know it, too. So they ... warn us to try to save ourselves because ... they can't help us any more."

There are many reasons why a man can feel shame that he belongs to a race which can do the things that some men do. But sometimes there are reasons to be proud, as well. The home world of this colony was doomed, but it sent a warning to the tiny group on the colony-world, to allow them to try to save themselves.

"I ... wish we were there to ... share what they have to face," said Riki. Her voice sounded as if her throat hurt. "I ... don't want to keep on living if ... everybody who ... ever cared about us is going to die!"

Massy felt lonely. He could understand that nobody would want to live as the only human alive. Nobody would want to live as a member of the only group of people left alive. And everybody thinks of his home planet as all the world there is. *I don't think that way*, thought Massy. *But maybe it's the way I'd feel about living if Riki were to die.* It would be natural to want to share any danger or any disaster she faced. Which he was.

"L-look!" he said, stammering a little "You don't see! It isn't a case of your living while they die! If your home world becomes like this, what will this be like? We're farther from the sun! We're colder to start with! Do you think we'll live through anything they can't take? Food supplies or no, equipment or no, do you think we've got a chance? Use your brains!"

Herndon and Riki stared at him. And then some of the strained look left Riki's face and body. Herndon blinked, and said slowly:

"Why ... that's so! We were thought to be taking a terrific risk when we came here. But it'll be as much worse here—Of course! We are in the same fix they're in!"

He straightened a little. Color actually came back into his face. Riki managed to smile. And then Herndon said almost naturally:

"That makes things look more sensible! We've got to fight for our lives, too! And we've very little chance of saving them! What do we do about it, Massy?"

II

The sun was halfway toward mid-sky, and still attended by its sun-dogs, though they were fainter than at the horizon. The sky was darker. The mountain peaks reached skyward, serene and utterly aloof from the affairs of men. This was a frozen world, where there should be no inhabitants. The city was a fleet of metal hulks, neatly arranged on the valley floor, emptied of the material they had brought for the building of the colony. At the upper end of the valley the landing-grid stood. It was a gigantic skeleton of steel, rising from legs of unequal length bedded in the hillsides, and reaching two thousand feet toward the stars. Human figures, muffled almost past recognition, moved about a catwalk three-quarters of the way up. There was a tiny glittering below where they moved. They were, of course, men using sonic ice-breakers to shatter the frost which formed on the framework at night. Falling shards of crystal made a liquidlike flashing. The landing-grid needed to be cleared every ten days or so. Left uncleared, it would acquire an increasingly thick coating of ice. In time it could collapse. But long before that time it would have ceased to operate, and without its operation there could be no space travel. Rockets for lifting spaceships were impossibly heavy, for practical use. But the landing-grids could lift them out to the unstressed space where Lawlor drives could work, and draw them to ground with cargoes they couldn't possibly have carried if they'd needed rockets.

Massy reached the base of the grid on foot. It was not far from the village of drone-hulls. He was dwarfed by the ground-level upright beams. He went

through the cold-lock to the small control-house at the grid's base.

He nodded to the man on standby as he got painfully out of his muffling garments.

"Everything all right?" he asked.

The standby operator shrugged. Massy was Colonial Survey. It was his function to find fault, to expose inadequacies in the construction and operation of colony facilities. *It's natural for me to be disliked by men whose work I inspect*, thought Massy. *If I approve it doesn't mean anything, and if I protest, it's bad*. He had always been lonely, but it was a part of the job.

"I think," he said painstakingly, "that there ought to be a change in maximum no-drain voltage. I'd like to check it."

The operator shrugged again. He pressed buttons under a phone-plate.

"Shift to reserve power," he commanded, when a face appeared in the plate. "Gotta check no-drain juice."

"What for?" demanded the face in the plate.

"You-know-who's got ideas," said the grid operator scornfully. "Maybe we've been skimping something. Maybe there's some new specification we didn't know about. Maybe anything! But shift to reserve power."

The face in the screen grumbled. Massy swallowed. It was not a Survey officer's privilege to maintain discipline. But there was no particular virtue in discipline here and now. He watched the current-demand dial. It stood a little above normal day-drain, which was understandable. The outside temperature was down. There was more power needed to keep the dwellings warm, and there was always a lot of power needed in the mine the colony had been formed to exploit. The mine had to be warmed for the men who worked to develop it.

The demand-needle dropped abruptly, and hung steady, and dropped again and again as additional parts of the colony's power-uses were switched to reserve. The needle hit bottom. It stayed there.

Massy had to walk around the standby man to get at the voltmeter. It was built around standard, old-fashioned vacuum tubes—standard for generations, now. Massy patiently hooked it up and warmed the tubes and tested it. He pushed in the contact-plugs. He read the no-drain voltage. He

licked his lips and made a note. He reversed the leads, so it would read backward. He took another reading. He drew in his breath very quietly.

"Now I want the power turned on in sections," he told the operator. "The mine first, maybe. It doesn't matter. But I want to get voltage-readings at different power take-offs."

The operator looked pained. He spoke with unnecessary elaboration to the face in the phone-plate, and grudgingly went through with the process by which Massy measured the successive drops in voltage with power drawn from the ionosphere. The current available from a layer of ionized gas is, in effect, the current-flow through a conductor with marked resistance. It is possible to infer a gas' ionization from the current it yields.

The cold-lock door opened. Riki Herndon came in, panting a little.

"There's another message from home," she said sharply. Her voice seemed strained. "They picked up our answering-beam and are giving the information you asked for."

"I'll be along," said Massy. "I just got some information here."

He got into his cold-garments again. He followed her out of the control-hut.

"The figures from home aren't good," said Riki evenly, when mountains visibly rose on every hand around them. "Ken says they're much worse than he thought. The rate of decline in the solar constant's worse than we figured or could believe."

"I see," said Massy, inadequately.

"It's absurd!" said Riki fiercely. "It's monstrous! There've been sunspots and sunspot cycles all along! I learned about them in school! I learned myself about a four-year and a seven-year cycle, and that there were others! They should have known! They should have calculated in advance! Now they talk about sixty-year cycles coming in with a hundred-and-thirty-year cycle to pile up with all the others—But what's the use of scientists if they don't do their work right and twenty million people die because of it?"

Massy did not consider himself a scientist, but he winced. Riki raged as they moved over the slippery ice. Her breath was an intermittent cloud about her shoulders. There was white frost on the front of her cold-garments.

He held out his hand quickly as she slipped, once.

"But they'll beat it!" said Riki in a sort of angry pride. "They're starting to build more landing-grids, back home. Hundreds of them! Not for ships to land by, but to draw power from the ionosphere! They figure that one ship-size grid can keep nearly three square miles of ground warm enough to live on! They'll roof over the streets of cities. Then they'll plant food-crops in the streets and gardens, and do what hydroponic growing they can. They are afraid they can't do it fast enough to save everybody, but they'll try!"

Massy clenched his hands inside their bulky mittens.

"Well?" demanded Riki. "Won't that do the trick?"

Massy said: "No."

"Why not?" she demanded.

"I just took readings on the grid, here. The voltage and the conductivity of the layer we draw power from, both depend on ionization. When the intensity of sunlight drops, the voltage drops and the conductivity drops, too. It's harder for less power to flow to the area the grid can tap—and the voltage-pressure is lower to drive it."

"Don't say any more!" cried Riki. "Not another word!"

Massy was silent. They went down the last small slope. They passed the opening of the mine—the great drift which bored straight into the mountain. They could look into it. They saw the twin rows of brilliant roof-lights going toward the heart of the stony monster.

They had almost reached the village when Riki said in a stifled voice:

"How bad is it?"

"Very," admitted Massy. "We have here the conditions the home planet will have in two hundred days. Originally we could draw less than a fifth the power they count on from a grid on Lani II."

Riki ground her teeth.

"Go on!" she said challengingly.

"Ionization here is down ten per cent," said Massy. "That means the voltage is down—somewhat more. A great deal more. And the resistance of the layer is greater. Very much greater. When they need power most, on the home planet, they won't draw more from a grid than we do now. It won't be enough."

They reached the village. There were steps to the cold-lock of Herndon's office-hull. They were ice-free, because like the village walkways they were warmed to keep frost from depositing on them. Massy made a mental note.

In the cold-lock, the warm air pouring in was almost stifling. Riki said defiantly:

"You might as well tell me now!"

"We could draw one-fifth as much power, here, as the same sized grid would yield on your home world," he said grimly. "We are drawing—call it sixty per cent of normal. A shade over one-tenth of what they must expect to draw when the real cold hits them. But their estimates are nine times too high." He said heavily, "One grid won't warm three square miles of city. About a third of one is closer. But—"

"That won't be the worst!" said Riki in a choked voice. "Is that right? How much good will a grid do?"

Massy did not answer.

The inner cold-lock door opened. Herndon sat at his desk, even paler than before, listening to the hash of noises that came out of the speaker. He tapped on the desktop, quite unconscious of the action. He looked almost desperately at Massy.

"Did she ... tell you?" he asked in a numb voice. "They hope to save maybe half the population. All the children anyhow—"

"They won't," said Riki bitterly.

"Better go transcribe the new stuff that's come in," said her brother dully. "We might as well know what it says."

Riki went out of the office. Massy laboriously shed his cold-garments. He said uncomfortably:

"The rest of the colony doesn't know what's up yet. The operator at the grid didn't, certainly. But they have to know."

"We'll post the messages on the bulletin board," said Herndon apathetically. "I wish I could keep it from them. It's not fun to live with. I ... might as well not tell them just yet."

"To the contrary," insisted Massy. "They've got to know right away! You're going to issue orders and they'll need to understand how urgent they are!"

Herndon looked absolutely hopeless.

"What's the good of doing anything?" When Massy frowned, he added as if exhausted: "Seriously, is there any use? You're all right. A Survey ship's due to take you away. It's not coming because they know there's something wrong, but because your job should be finished about now. But it can't do any good! It would be insane for it to land at home. It couldn't carry away more than a few dozen refugees, and there are twenty million people who're going to die. It might offer to take some of us. But ... I don't think many of us would go. I wouldn't. I don't think Riki would."

"I don't see—"

"What we've got right here," said Herndon, "is what they're going to have back home. And worse. But there's no chance for us to keep alive here! You are the one who pointed it out! I've been figuring, and the way the solar-constant curve is going—I plotted it from the figures they gave us—it couldn't possibly level out until the oxygen, anyhow, is frozen out of the atmosphere here. We aren't equipped to stand anything like that, and we can't get equipped. There couldn't be equipment to let us stand it indefinitely! Anyhow the maximum cold conditions will last two thousand days back home—six Earth-years. And there'll be storage of cold in frozen oceans and piled-up glaciers—It'll be twenty years before home will be back to normal in temperature, and the same here. Is there any point in trying to live—just barely to survive—for twenty years before there'll be a habitable planet to go back to?"

Massy said irritably:

"Don't be a fool! Doesn't it occur to you that this planet is a perfect experiment-station, two hundred days ahead of the home world, where ways

to beat the whole business can be tried? If we can beat it here, they can beat it there!"

Herndon said detachedly:

"Can you name one thing to try here?"

"Yes," snapped Massy. "I want the walk-heaters and the step-heaters outside turned off. They use power to keep walkways clear of frost and doorsteps not slippery. I want to save that heat!"

Herndon said without interest:

"And when you've saved it, what will you do with it?"

"Put it underground to be used as needed!" Massy said angrily. "Store it in the mine! I want to put every heating-device we can contrive to work in the mine! To heat the rock! I want to draw every watt the grid will yield and warm up the inside of the mountain while we can draw power to do it with! I want the deepest part of the mine too hot to enter! We'll lose a lot of heat, of course. It's not like storing electric power! But we can store heat now, and the more we store the more will be left when we need it!"

Herndon thought heavily. Presently he stirred slightly.

"Do you know, that is an idea—" He looked up. "Back home there was a shale-oil deposit up near the icecaps. It wasn't economical to mine it. So they put heaters down in bore-holes and heated up the whole shale deposit! Drill-holes let out the hot oil vapors to be condensed. They got out every bit of oil without disturbing the shale! And then ... why ... the shale stayed warm for years. Farmers bulldozed soil over it and raised crops with glaciers all around them! That could be done again. They could be storing up heat back home!"

Then he drooped.

"But they can't spare power to warm up the ground under cities. They need all the power they've got to build roofs. And it takes time to build grids."

Massy snapped:

"Yes, if they're building regulation ones! By the time they were finished they'd be useless! The ionization here is dropping already. But they don't need to build grids that will be useless later! They can weave cables together on the ground and hang them in the air by helicopters! They wouldn't hold up a landing ship for an instant, but they'll draw power right away! They'll even power the helis that hold them up! Of course they've defects! They'll have to come down in high winds. They won't be dependable. But they can put heat in the ground to come out under roofs, to grow food by, to save lives by. What's the matter with them?"

Herndon stirred again. His eyes ceased to be dull and lifeless.

"I'll give the orders for turning off the sidewalks. And I'll send what you just said back home. They ... should like it."

He looked very respectfully at Massy.

"I guess you know what I'm thinking right now," he said awkwardly.

Massy flushed. It was not dignified for a Colonial Survey officer to show off. He felt that Herndon was unduly impressed. But Herndon didn't see that the device wouldn't solve anything. It would merely postpone the effects of a disaster. It could not possibly prevent them.

"It ought to be done," he said curtly. "There'll be other things to be done, too."

"When you tell them to me," said Herndon warmly, "they'll get done! I'll have Riki put this into that pulse-code you explained to us and she'll get it off right away!"

He stood up.

"I didn't explain the code to her!" insisted Massy. "She was already translating it when you gave her my suggestion!"

"All right," said Herndon. "I'll get this sent back at once!"

He hurried out of the office. *This*, thought Massy irritably, *is how reputations are made, I suppose. I'm getting one.* But his own reaction was extremely inappropriate. If the people of Lani II did suspend helicopter-supported grids of wire in the atmosphere, they could warm masses of underground rock and stone and earth. They could establish what were practically reservoirs of life-

giving heat under their cities. They could contrive that the warmth from below would rise only as it was needed. But—

Two hundred days to conditions corresponding to the colony-planet. Then two thousand days of minimum-heat conditions. Then very, very slow return to normal temperature, long after the sun was back to its previous brilliance. They couldn't store enough heat for so long. It couldn't be done. It was ironic that in the freezing of ice and the making of glaciers the planet itself could store cold.

And there would be monstrous storms and blizzards on Lani II as it cooled. As cold conditions got worse the wire grids could be held aloft for shorter and shorter periods, and each time they would pull down less power than before. Their effectiveness would diminish even faster than the need for effectiveness increased.

Massy felt even deeper depression as he worked out the facts. His proposal was essentially futile. It would be encouraging, and to a very slight degree and for a certain short time it would palliate the situation on the inner planet. But in the long run its effect would be zero.

He was embarrassed, too, that Herndon was so admiring. Herndon would tell Riki that he was marvelous. She might—though cagily—be inclined to agree. But he wasn't marvelous. This trick of a flier-supported grid was not new. It had been used on Saril to supply power for giant peristaltic pumps emptying a polder that had been formed inside a ring of indifferently upraised islands.

All I know, thought Massy bitterly, is what somebody's showed me or I've read in books. And nobody's showed or written how to handle a thing like this!

He went to Herndon's desk. Herndon had made a new graph on the solar-constant observations forwarded from home. It was a strictly typical curve of the results of coinciding cyclic changes. It was the curve of a series of frequencies at the moment when they were all precisely in phase. From this much one could extrapolate and compute—

Massy took a pencil, frowning unhappily. His fingers clumsily formed equations and solved them. The result was just about as bad as it could be. The change in brightness of the sun Lani would not be enough to be observed on Kent IV—the nearest other inhabited world—when the light reached there four years from now. Lani would never be classed as a variable star, because the total change in light and heat would be relatively minute. But the formula for computing planetary temperatures is not simple. Among its factors are squares and cubes of the variables. Worse, the heat radiated from a sun's photosphere varies not as the square or cube, but as the fourth power of its absolute temperature. A very small change in the sun's effective temperature, producible by sunspots, could make an altogether disproportionate difference in the warmth its worlds received.

Massy's computations were not pure theory. The data came from Sol itself, where alone in the galaxy there had been daily solar-constant measurements for three hundred years. The rest of his deductions were based ultimately on Earth observations, too. Most scientific data had to refer back to Earth to get an adequate continuity. But there was no possible doubt about the sunspot data, because Sol and Lani were of the same type and nearly equal size.

Using the figures on the present situation, Massy reluctantly arrived at the fact that here, on this already-frozen world, the temperature would drop until CO₂ froze out of the atmosphere. When that happened, the temperature would plummet until there was no really significant difference between it and that of empty space. It is carbon dioxide which is responsible for the greenhouse effect, by which a planet is in thermal equilibrium only at a temperature above its surroundings—as a greenhouse in sunlight is warmer than the outside air.

The greenhouse effect would vanish soon on the colony-world. When it vanished on the mother planet—

Massy found himself thinking, *If Riki won't leave when the Survey ship comes, I'll resign from the Service. I'll have to if I'm to stay. And I won't go unless she does.*

"If you want to come, it's all right," said Massy ungraciously.

He waited while Riki slipped into the bulky cold-garments that were needed out-of-doors in the daytime, and were doubly necessary at night. There were heavy boots with inches-thick insulating soles, made in one piece with the many-layered trousers. There was the air-puffed, insulated over-tunic with its hood and mittens which were a part of the sleeves.

"Nobody goes outside at night," she said when they stood together in the cold-lock.

"I do," he told her. "I want to find out something."

The outer door opened and he stepped out. He held his arm for her, because the steps and walkway were no longer heated. Now they were covered with a filmy layer of something which was not frost, but a faint, faint bloom of powder. It was the equivalent of dust, but it was microscopic snow-crystals frozen out of the air by the unbearable chill of night.

There was no moon, of course, yet the ice-clad mountains glowed faintly. The drone-hulls arranged in such an orderly fashion were dark against the frosted ground. There was silence: stillness: the feeling of ancient quietude. No wind stirred anywhere. Nothing moved. Nothing lived. The soundlessness was enough to crack the eardrums.



Massy threw back his head and gazed at the sky for a very long time. Nothing. He looked down at Riki.

"Look at the sky," he commanded.

She raised her eyes. She had been watching him. But as she gazed upward she almost cried out. The sky was filled with stars in innumerable variety. But the brighter ones were as stars had never been seen before. Just as the sun in daylight had been accompanied by its sun-dogs—pale phantoms of itself ranged about it—so the brighter distant suns now shone from the center of rings of their own images. They no longer had the look of random placing. Those which were most distinct were patterns in themselves, and one's eyes strove instinctively to grasp the greater pattern in which such seeming artifacts must belong.

"Oh ... beautiful!" cried Riki softly, yet almost afraid.

"Look!" he insisted. "Keep looking!"

She continued to gaze, moving her eyes about hopefully. It was such a sight as no one could have imagined. Every tint and every color; every possible degree of brightness appeared. And there were groups of stars of the same brilliance which almost made triangles, but not quite. There were rose-tinted stars which almost formed an arc, but did not. And there were arrays which were almost lines and nearly formed squares and polygons, but never actually achieved them.

"It's ... beautiful!" said Riki breathlessly. "But what must I look for?"

"Look for what isn't there," he ordered.

She looked, and the stars were unwinking, but that was not extraordinary. They filled all the firmament, without the least space in which some tiny sparkle of light was not to be found. But that was not remarkable, either. Then there was a vague flickering grayish glow somewhere indefinite. It vanished. Then she realized.

"There's no aurora!" she exclaimed.

"That's it," said Massy. "There've always been auroras here. But no longer. We may be responsible. I wish I thought it wise to turn everything back to reserve power for a while. We could find out. But we can't afford it. There was just the faintest possible gray flickering just now. But there ought to be armies of light marching across the sky. The aurora here—it was never missing! But it's gone now."

"I ... looked at it when we first landed," admitted Riki. "It was unbelievable! But it was terribly cold, out of shelter. And it happened every night, so I said to myself I'd look tomorrow, and then tomorrow again. So it got so I never looked at all."

Massy kept his eyes where the faint gray flickering had been. And once one realized, it was astonishing that the former nightly play of ghostly colors should be absent.

"The aurora," he said dourly, "happens in the very upper limits of the air ... fifty ... seventy ... ninety miles up, when God-knows-what emitted particles from the sun come streaking in, drawn by the planet's magnetic field. The aurora's a phenomenon of ions. We tap the ionosphere a long way down from where it plays, but I'm wondering if we stopped it."

"We?" said Riki, shocked. "We—humans?"

"We tap the ions of their charges," he said somberly, "that the sunlight made by day. We're pulling in all the power we can. I wonder if we've drained the aurora of its energy, too."

Riki was silent. Massy gazed, still searching. But he shook his head.

"It could be," he said in a carefully detached voice. "We didn't draw much power by comparison with the amount that came. But the ionization is an ultraviolet effect. Atmospheric gases don't ionize too easily. After all, if the solar constant dropped a very little, it might mean a terrific drop in the ultraviolet part of the spectrum—and that's what makes ions of oxygen and nitrogen and hydrogen and such. The ion-drop could easily be fifty times as great as the drop in the solar constant. And we're drawing power from the little that's left."

Riki stood very still. The cold was horrible. Had there been a wind, it could not have been endured for an instant. But the air was motionless. Yet its coldness was so great that the inside of one's nostrils ached, and the inside of one's chest was aware of chill. Even through the cold-garments there was the feeling as of ice without.

"I'm beginning," said Massy, "to suspect that I'm a fool. Or maybe I'm an optimist. It might be the same thing. I could have guessed that the power we could draw would drop faster than our need for power increased. If we've drained the aurora of its light, we're scraping the bottom of the barrel. And it's a shallower barrel than one would suspect."

There was stillness again. Riki stood mousy-quiet. *When she realizes what this means, thought Massy grimly, she won't admire me so much. Her brother's built me up. But I've been a fool, figuring out excuses to hope. She'll see it.*

"I think," said Riki quietly, "that you're telling me that after all we can't store up heat to live on, down in the mine."

"We can't," agreed Massy grimly. "Not much, nor long. Not enough to matter."

"So we won't live as long as Ken expects?"

"Not nearly as long," said Massy evenly. "He's hoping we can find out things to be useful back on Lani II. But we'll lose the power we can get from our grid long before even their new grids are useless. We'll have to start using our reserve power a lot sooner. It'll be gone—and us with it—before they're really in straits for living-heat."

Riki's teeth began to chatter.

"This sounds like I'm scared," she said angrily, "but I'm not! I'm just freezing! If you want to know, I'd a lot rather have it the way you say! I won't have to grieve over anybody, and they'll be too busy to grieve for me! Let's go inside while it's still warm!"

He helped her back into the cold-lock, and the outer door closed. She was shivering uncontrollably when the warmth came pouring in.

They went into Herndon's office. He came in as Riki was peeling off the top part of her cold-garments. She still shivered. He glanced at her and said to Massy:

"There's been a call from the grid-control shack. It looks like there's something wrong, but they can't find anything. The grid is set for maximum power-collection, but it's bringing in only fifty thousand kilowatts!"

"We're on our way back to savagery," said Massy, with an attempt at irony.

It was true. A man can produce two hundred and fifty watts from his muscles for a reasonable length of time. When he has no more power, he is a savage. When he gains a kilowatt of energy from the muscles of a horse, he is a barbarian—but the new power cannot be directed wholly as he wills. When he can apply it to a plow he has high barbarian culture, and when he adds still more he begins to be civilized. Steam power put as much as four kilowatts to work for every human being in the first industrialized countries, and in the mid-twentieth century there was sixty kilowatts per person in the more advanced nations. Nowadays, of course, a modern culture assumed five hundred as a minimum. But there was less than half that in the colony on Lani II. And its environment made its own demands.

"There can't be any more," said Riki, trying to control her shivering. "We're even using the aurora and there isn't any more power. It's running out. We'll go even before the people at home, Ken."

Herndon's features looked very pinched.

"But we can't! We mustn't!" He turned to Massy. "We do them good, back home! There was panic. Our report about cable-grids has put heart in people. They're setting to work—magnificently! So we're some use! They know we're worse off than they are, and as long as we hold on they'll be encouraged! We've got to keep going somehow!"

Riki breathed deeply until her shivering stopped. Then she said calmly:

"Haven't you noticed, Ken, that Mr. Massy has the viewpoint of his profession? His business is finding things wrong with things. He was deposited in our midst to detect defects in what we did and do. He has the habit of looking for the worst. But I think he can turn the habit to good use. He did turn up the idea of cable-grids."

"Which," said Massy, "turns out to be no good at all. They'd be some good if they weren't needed, really. But the conditions that make them necessary make them useless!"

Riki shook her head.

"They are useful!" she said firmly. "They're keeping people at home from despairing. Now, though, you've got to think of something else. If you think of enough things, one will do good the way you want—more than making people feel better."

"What does it matter how people feel?" he demanded bitterly. "What difference do feelings make? Facts are facts! One can't change facts!"

Riki said with no less firmness:

"We humans are the only creatures in the universe who don't do anything else! Every other creature accepts facts. It lives where it is born, and it feeds on the food that is there for it, and it dies when the facts of nature require it to. We humans don't. Especially we women! We won't let men do it, either! When we don't like facts—mostly about ourselves—we change them. But important facts we disapprove of—we ask men to change for us. And they do!"

She faced Massy. Rather incredibly, she grinned at him.

"Will you please change the facts that look so annoying just now, please? Please?" Then she elaborately pantomimed an over-feminine girl's look of wide-eyed admiration. "You're so big and strong! I just know you can do it—for me!"

She abruptly dropped the pretense and moved toward the door. She half-turned then, and said detachedly:

"But about half of that is true."

The door slid shut behind her. Massy thought bitterly, *Her brother admires me. She probably thinks I really can do something!* It suddenly occurred to him that she knew a Colonial Survey ship was due to stop by here to pick him up. She believed he expected to be rescued, even though the rest of the colony could not be, and most of it wouldn't consent to leave their kindred when the death of mankind in this solar system took place. He said awkwardly:

"Fifty thousand kilowatts isn't enough to land a ship."

Herndon frowned. Then he said:

"Oh. You mean the Survey ship that's to pick you up can't land? But it can go in orbit and put down a rocket landing-boat for you."

Massy flushed.

"I wasn't thinking of that. I'd something more in mind. I ... rather like your sister. She's ... pretty wonderful. And there are some other women here in the colony, too. About a dozen all told. As a matter of self-respect I think we ought to get them away on the Survey ship. I agree that they wouldn't consent to go. But if they had no choice—if we could get them on board the grounded ship, and they suddenly found themselves ... well ... kidnaped and outward-bound not by their own fault.... They could be faced with the accomplished fact that they had to go on living."

Herndon said evenly:

"That's been in the back of my mind for some time. Yes. I'm for that. But if the Survey ship can't land—"

"I believe I can land it regardless," said Massy doggedly. "I can find out, anyhow. I'll need to try things. I'll need help ... work done. But I want your promise that if I can get the ship to ground you'll conspire with her skipper and arrange for them to go on living."

Herndon looked at him.

"Some new stuff—in a way," said Massy uncomfortably. "I'll have to stay aground to work it. It's also part of the bargain that I shall. And, of course, your sister can't know about it, or she can't be fooled into living."

Herndon's expression changed a little.

"What'll you do? Of course it's a bargain."

"I'll need some metals we haven't smelted so far," said Massy. "Potassium if I can get it, sodium if I can't, and at worst I'll settle for zinc. Cesium would be best, but we've found no traces of it."

Herndon said thoughtfully:

"No-o-o. I think I can get you sodium and potassium, from rocks. I'm afraid no zinc. How much?"

"Grams," said Massy. "Trivial quantities. And I'll need a miniature landing-grid built. Very miniature."

Herndon shrugged his shoulders.

"It's over my head. But just to have work to do will be good for everybody. We've been feeling more frustrated than any other humans in history. I'll go round up the men who'll do the work. You talk to them."

The door closed behind him. Massy very deliberately got out of his cold-clothing. He thought, *She'll rave when she finds her brother and I have deceived her.* Then he thought of the other women. *If any of them are married, we'll have to see if there's room for their husbands. I'll have to*

dress up the idea. Make it look like reason for hope, or the women would find out. But not many can go—

He knew very closely how many extra passengers could be carried on a Survey ship, even in such an emergency as this. Living quarters were not luxurious, at best. Everything was cramped and skimped. Survey ships were rugged, tiny vessels which performed their duties amid tedium and discomfort and peril for all on board. But they could carry away a very few unwilling refugees to Kent IV.

He settled down at Herndon's desk to work out the thing to be done.

It was not unreasonable. Tapping the ionosphere for power was something like pumping water out of a pipe-well in sand. If the water-table was high, there was pressure to force the water to the pipe, and one could pump fast. If the water-table were low, water couldn't flow fast enough. The pump would suck dry. In the ionosphere, the level of ionization was at once like the pressure and the size of the sand-grains. When the level was high, the flow was vast because the sand-grains were large and the conductivity high. But as the level lessened, so did the size of the sand-grains. There was less to draw, and more resistance to its flow.

But there had been one tiny flicker of auroral light over by the horizon. There was still power aloft. If Massy could in a fashion prime the pump: if he could increase the conductivity by increasing the ions present around the place where their charges were drawn away—why—he could increase the total flow. It would be like digging a brick-well where a pipe-well had been. A brick-well draws water from all around its circumference.

So Massy computed carefully. It was ironic that he had to go to such trouble simply because he didn't have test-rockets like the Survey uses to get a picture of a planet's weather-pattern. They rise vertically for fifty miles or so, trailing a thread of sodium-vapor behind them. The trail is detectable for some time, and ground-instruments record each displacement by winds blowing in different directions at different speeds, one over the other. Such a rocket with its loading slightly changed would do all Massy had in mind. But he didn't have one, so something much more elaborate was called for.

She'll think I'm clever, he reflected wryly, but all I'm doing is what I've been taught. I wouldn't have to work it out if I had a rocket.

Still, there was some satisfaction in working out this job. A landing-grid has to be not less than half a mile across and two thousand feet high because its field has to reach out five planetary diameters to handle ships that land and take off. To handle solid objects it has to be accurate—though power can be drawn with an improvisation. To thrust a sodium-vapor bomb anywhere from twenty to fifty miles high—why—he'd need a grid only six feet wide and five high. It could throw much higher, of course. It could hold, at that. But doubling the size would make accuracy easier.

He tripled the dimensions. There would be a grid eighteen feet across and fifteen high. Tuned to the casing of a small bomb, it could hold it steady at seven hundred and fifty thousand feet—far beyond necessity. He began to make the detail drawings.

Herndon came back with half a dozen chosen colonists. They were young men, technicians rather than scientists. Some of them were several years younger than Massy. There were grim and stunned expressions on some faces, but one tried to pretend nonchalance, and two seemed trying to suppress fury at the monstrous occurrence that would destroy not only their own lives, but everything they remembered on the planet which was their home. They looked almost challengingly at Massy.

He explained. He was going to put a cloud of metallic vapor up in the ionosphere. Sodium if he had to, potassium if he could, zinc if he must. Those metals were readily ionized by sunlight—much more readily than atmospheric gases. In effect, he was going to supply a certain area of the ionosphere with material to increase the efficiency of sunshine in providing electric power. As a sideline, there would be increased conductivity from the normal ionosphere.

"Something like this was done centuries ago, back on Earth," he explained carefully. "They used rockets, and made sodium-vapor clouds as much as twenty and thirty miles long. Even nowadays the Survey uses test-rockets with trails of sodium-vapor. It will work to some degree. We'll find out how much."

He felt Herndon's eyes upon him. They were almost dazedly respectful. But one of the technicians said coldly:

"How long will those clouds last?"

"That high, three or four days," Massy told him. "They won't help much at night, but they should step up power-intake while the sun shines on them."

A man in the back said crisply:

"Hup!" The significance was, "Let's go!" Then somebody said feverishly, "What do we do? Got working drawings? Who makes the bombs? Who does what? Let's get at this!"

Then there was confusion, and Herndon had vanished. Massy suspected he'd gone to have Riki put this theory into dot-and-dash code for beam-transmission back to Lani II. But there was no time to stop him. These men wanted precise information, and it was half an hour before the last of them had gone out with free-hand sketches, and had come back for further explanation of a doubtful point, and other men had come in hungrily to demand a share in the job.

When he was alone again, Massy thought, *Maybe it's worth doing because it'll get Riki on the Survey ship. But they think it means saving the people back home!*

Which it didn't. Taking energy out of sunlight is taking energy out of sunlight, no matter how you do it. Take it out as electric power, and there's less heat left. Warm one place with electric power, and everywhere else is a little colder. There's an equation. On this colony-world it wouldn't matter, but on the home world it would. The more there was trickery to gather heat, the more heat was needed. Again it might postpone the death of twenty million people, but it would never, never, never prevent it.

The door slid aside and Riki came in. She stammered a little.

"I ... just coded what Ken told me to send back home. It will ... it will do everything! It's wonderful! I ... wanted to tell you!"

Massy writhed internally. It wasn't wonderful.

"Consider," he said in a desperate attempt to take it lightly, "consider that I've taken a bow."

He tried to smile. It was not a success. And Riki suddenly drew a deep breath and looked at him in a new fashion.

"Ken's right," she said softly. "He says you can't get conceited. You're not satisfied with yourself even now, are you?" She smiled, rather gravely. Then she said, "But what I like is that you aren't really smart. A woman can make you do things. I have!"

He looked at her uneasily. She grinned.

"I, even I, can at least pretend to myself that I help bring this about! If I hadn't said please change the facts that are so annoying, and if I hadn't said you were big and strong and clever—I'm going to tell myself for the rest of my life that I helped make you do it!"

Massy swallowed.

"I'm afraid," he said miserably, "that it won't work again."

She cocked her head on one side.

"No?"

He stared at her apprehensively. And then with a bewildering change of emotional reaction, he saw that her eyes were filled with tears. She stamped her foot.

"You're ... horrible!" she cried. "Here I come in, and ... and if you think you can get me kidnaped to safety ... without even telling me that you 'rather like' me, like you told my brother, or that 'I'm pretty wonderful'—If you think."

He was stunned, that she knew. She stamped her foot again.

"For Heaven's sake!" she wailed. "Do I have to *ask* you to kiss me?"

IV

During the last night of preparation, Massy sat by a thermometer registering the outside temperature. He hovered over it as one might over a sick child.

He watched it and sweated, though the inside temperature of the drone-hull was lowered to save power. There was nothing he could actually do. At midnight the thermometer said it was seventy degrees below zero Fahrenheit. At halfway to dawn it was eighty degrees below zero Fahrenheit. The hour before dawn it was eighty-five degrees below zero. Then he sweated profusely. The meaning of the slowed descent was that carbon dioxide was being frozen out of the upper layers of the atmosphere. The frozen particles were drifting slowly downward, and as they reached lower and faintly warmer levels they returned to the state of gas. But there was a level, above the CO₂, where the temperature was plummeting.

The height to which carbon dioxide existed was dropping—slowly, but inexorably. And above the carbon-dioxide level there was no bottom limit to the temperature. The greenhouse effect was due to CO₂. Where it wasn't, the cold of space moved down. If at ground-level the thermometer read ever so slightly lower than one hundred and nine below zero—why—everything was finished. Without the greenhouse effect, the night-side of the planet would lose its remaining heat with a rush. Even the day-side, once cold enough, would lose heat to emptiness as fast as it came from the sun. Minus one hundred and nine point three was the critical reading. If it went down to that, it would plunge to a hundred and fifty—two hundred degrees below zero! And it would never come up again.

There would be rain at nightfall—a rain of oxygen frozen to a liquid and splashing on the ground. Human life would be quite simply impossible, in any shelter and under any conditions. Even spacesuits would not protect against an atmosphere sucking heat from it at that rate. A spacesuit can be heated against the loss of temperature due to radiation in a vacuum. It could not be heated against nitrogen which would chill it irresistibly by contact.

But, as Massy sweated over it, the thermometer steadied at minus eighty-five degrees. When the dawn came, it rose to seventy. By mid-morning, the temperature in bright sunshine was no lower than sixty-five degrees below zero.

But there was no bounce left in Massy when Herndon came for him.

"Your phone-plate's been flashing," said Herndon, "and you didn't answer. Must have had your back to it. Riki's over in the mine, watching them get

things ready. She was worried that she couldn't call you. Asked me to find out what was the trouble."

Massy said heavily:

"Has she got something to heat the air she breathes?"

"Naturally," said Herndon. He added curiously, "What's the matter?"

"We almost took our licking," Massy told him. "I'm afraid for tonight, and tomorrow night, too. If the CO₂ freezes—"

"We'll have power!" Herndon insisted. "We'll build ice tunnels and ice domes. We'll build a city under ice, if we have to. But we'll have power. We'll be all right!"

"I doubt it very much," said Massy. "I wish you hadn't told Riki of the bargain to get her away from here when the Survey ship comes!"

Herndon grinned.

"Is the little grid ready?" asked Massy.

"Everything's set," said Herndon exuberantly. "It's in the mine-tunnel with radiant heaters playing on it. The bombs are ready. We made enough to last for months, while we were at it. No use taking chances!"

Massy looked at him queerly. Then he said:

"We might as well go out and try the thing, then."

But he was very tired. He was not elated. *Riki can't be gotten away*, he thought wearily, *and I'm not going to go because it isn't quite fitting to go and leave her. They'll all be rejoicing presently, but nothing's settled.* Then he thought with exquisite irony, *She thinks I was inspired to genius by her, when I haven't done a thing I wasn't taught or didn't get out of books!*

He put on the cold-garments as they were now modified for the increased frigidity. Nobody could breathe air at minus sixty-five degrees without getting his lungs frost-bitten. So there was now a plastic mask to cover one's face, and the air one breathed outdoors was heated as it came through a wire-

gauze snout. But still it was not wise to stay out of shelter for too long a time.

Massy went out-of-doors. He stepped out of the cold-lock and gazed about him. The sun seemed markedly paler, and now it had lost its sun-dogs again. Ice crystals no longer floated in the almost congealed air. The sky was dark. It was almost purple, and it seemed to Massy that he could detect faint flecks of light in it. They would be stars, shining in the daytime.

There seemed to be no one about at all, only the white coldness of the mountains. But there was a movement at the mine-drift, and something came out of it. Four men appeared, muffled up like Massy himself. They rolled the eighteen-foot grid out of the mine-mouth, moving it on those inflated bags which are so much better than rollers for rough terrain. They looked absurdly like bears with steaming noses, in their masks and clothing. They had some sort of powered pusher with them, and they got the metal cage to the very top of a singularly rounded stone upcrop which rose in the center of the valley.

"We picked that spot," said Herndon's muffled voice through the chill, "because by shifting the grid's position it can be aimed, and be on a solid base. Right?"

"Quite all right," said Massy. "We'll go work it."

He moved heavily across the valley, in which nothing moved except the padded figures of the four technicians. Their wire-gauze breathing-masks seemed to emit smoke. They waved to him in greeting.

I'm popular again, he thought drearily, but it doesn't matter. Getting the Survey ship to ground won't help now, since Riki's forewarned. And this trick won't solve anything permanently on the home planet. It'll just postpone things.

He had a very peculiar ache inside. A Survey officer is naturally lonely. Massy had been lonely before he even entered the Service. He hadn't had a feeling of belonging anywhere, or with anyone, and no planet was really his home. Now he could believe that he belonged with someone. But there was the slight matter of a drop in the solar constant of an unimportant sol-type sun, and nothing could come of it.

Even when Riki—muffled like the rest—waved to him from the mouth of the tunnel, his spirits did not lift. The thing he wanted was to look forward to years and years of being with Riki. He wanted, in fact, to look forward to forever. And there might not be a tomorrow.

"I had the control board rolled out here," she called breathlessly through her mask. "It's cold, but you can watch!"

It wouldn't be much to watch. If everything went all right, some dial-needles would kick over violently, and their readings would go up and up. But they wouldn't be readings of temperature. Presently the big grid would report increased power from the sky. But tonight the temperature would drop a little farther. Tomorrow night it would drop farther still. When it reached one hundred and nine point three degrees below zero at ground-level—why it would keep on falling indefinitely. Then it wouldn't matter how much power could be drawn from the sky. The colony would die.

One of the figures that looked like a bear now went out of the mine-mouth, trudging toward the grid. It carried a muffled, well-wrapped object in its arms. It stooped and crept between the spokes of the grid. It put the object on the stone. Massy traced cables with his eyes. From the grid to the control board. From the board back to the reserve-power storage cells, deep in the mountain.

"The grid's tuned to the bomb," said Riki breathlessly, close beside him. "I checked that myself!"

The bearlike figure out in the valley jerked at the bomb. There was a small rising cloud of grayish vapor. It continued. The figure climbed hastily out of the grid. When the man was clear, Massy threw a switch.

There was a very tiny whining sound, and the wrapped, ridiculously smoking object leaped upward. It seemed to fall toward the sky. There was no more of drama than that. An object the size of a basketball fell upward, swiftly, until it disappeared. That was all.

Massy sat quite still, watching the control-board dials. Presently he corrected this, and shifted that. He did not want the bomb to have too high an upward

velocity. At a hundred thousand feet it would find very little air to stop the rise of the vapor it was to release.

The field-focus dial reached its indication of one hundred thousand feet. Massy reversed the lift-switch. He counted and then switched the power off. The small, thin whine ended.

He threw the power-intake switch, which could have been on all the time. The power-yield needle stirred. The minute grid was drawing power like its vaster counterpart. But its field was infinitesimal by comparison. It drew power as a soda straw might draw water from wet sand.

Then the intake-needle kicked. It swung sharply, and wavered, and then began a steady, even, climbing movement across the markings on the dial-face. Riki was not watching that.

"They see something!" she panted. "Look at them!"

The four men who had trundled the smaller grid to its place, now stared upward. They flung out their arms. One of them jumped up and down. They leaped. They practically danced.

"Let's go see," said Massy.

He went out of the tunnel with Riki. They gazed upward. And directly overhead, where the sky was darkest blue and where it had seemed that stars shone through the daylight—there was a cloud. It seemed to Massy, very quaintly, that it was no bigger than a man's hand. But it grew. Its edges were yellow—saffron-yellow. It expanded and spread. Presently it began to thin. As it thinned, it began to shine. It was luminous. And the luminosity had a strange, familiar quality.

Somebody came panting down the tunnel, from inside the mountain.

"The grid—" he panted. "The big grid! It's ... pumping power! Big power! BIG power!"

He went pounding back, to gaze raptuously at the new position of a thin black needle on a large white dial, and to make incoherent noises of rejoicing as it moved very, very slowly toward higher and ever higher readings.

But Massy looked puzzledly at the sky, as if he did not quite believe his eyes. The cloud now expanded very slowly, but still it grew. And it was not

regular in shape. The bomb had not shattered quite evenly, and the vapor had poured out more on one side than the other. There was a narrow, arching arm of brightness—

"It looks," said Riki breathlessly, "like a comet!"

And then Massy froze in every muscle. He stared at the cloud he had made aloft, and his hands clenched in their mittens, and he swallowed convulsively behind his cold-mask.

"Th-that's it," he said in a very queer voice indeed. "It's ... very much like a comet. I'm glad you said that! We can make something even more like a comet. We ... we can use all the bombs we've made, right away, to make it. And we've got to hurry so it won't get any colder tonight!"

Which, of course, sounded like insanity. Riki looked apprehensively at him. But Massy had just thought of something. And nobody had taught it to him and he hadn't gotten it out of books. But he'd seen a comet.

The new idea was so promising that he regarded it with anguished unease for fear it would not hold up. It was an idea that really ought to change the facts resulting naturally from a lowered solar constant in a sol-type star.

Half the colony set to work to make more bombs when the effect of the second bomb showed up. They were not very efficient, at first, because they tended to want to stop work and dance, from time to time. But they worked with an impassioned enthusiasm. They made more bomb-casings, and they prepared more sodium and potassium metal and more fuses, and more insulation to wrap around the bombs to protect them from the cold of airless space.



Because these were to go out to airlessness. The miniature grid could lift and hold a bomb steady in its field-focus at seven hundred and fifty thousand feet. But if a bomb was accelerated all the way out to that point, and the field was then snapped off—Why, it wasn't held anywhere. It kept on going with its attained velocity. And it burst when its fuse decided that it should, whereupon immediately a mass of sodium and potassium vapor, mixed with the fumes of high explosive, flung itself madly in all directions, out between the stars. Absolute vacuum tore the compressed gasified metals apart. The separate atoms, white-hot from the explosion, went swirling through sunlit space. The sunlight was dimmed a trifle, to be sure. But individual atoms of the lighter alkaline-earth metals have marked photo-electric properties. In

sunshine these gas-molecules ionized, and therefore spread more widely, and did not coalesce into even microscopic droplets.

They formed, in fact, a cloud in space. An ionized cloud, in which no particle was too large to be responsive to the pressure of light. The cloud acted like the gases of a comet's tail. It was a comet's tail, though there was no comet. And it was an extraordinary comet's tail because it is said that you can put a comet's tail in your hat, at normal atmospheric pressure. But this could not have been put in a hat. Even before it turned to gas, it was the size of a basketball. And, in space, it glowed.

It glowed with the brightness of the sunshine on it, which was light that would normally have gone away through the interstellar dark. And it filled one corner of the sky. Within one hour it was a comet's tail ten thousand miles long, which visibly brightened the daytime heavens. And it was only the first of such reflecting clouds.

The next bomb set for space exploded in a different quarter, because Massy'd had the miniature grid wrestled around the upcrop to point in a new and somewhat more carefully chosen line. The third bomb spattered brilliance in a different section still. And the brilliance lasted.

Massy flung his first bombs recklessly, because there could be more. But he was desperately anxious to hang as many comet tails as possible around the colony-planet before nightfall. He didn't want it to get any colder.

And it didn't. In fact, there wasn't exactly any real nightfall on Lani III that night.

The planet turned on its axis, to be sure. But around it, quite close by, there hung gigantic streamers of shining gas. At their beginning, those streamers bore a certain resemblance to the furry wild-animal tails that little boys like to have hanging down from hunting-caps. Only they shone. And as they developed they merged, so that there was an enormous shining curtain about Lani III. There were draperies of metal-mist to capture sunlight that should have been wasted, and to diffuse very much of it to Lani II. At midnight there was only one spot in all the night-sky where there was really darkness. That was directly overhead—directly outward from the planet from the sun. Gigantic shining streamers formed a wall, a tube, of comet-tail material, yet many times more dense and therefore brighter—which shielded the colony-

world against the dark and cold, and threw upon it a brilliant, warming brightness.

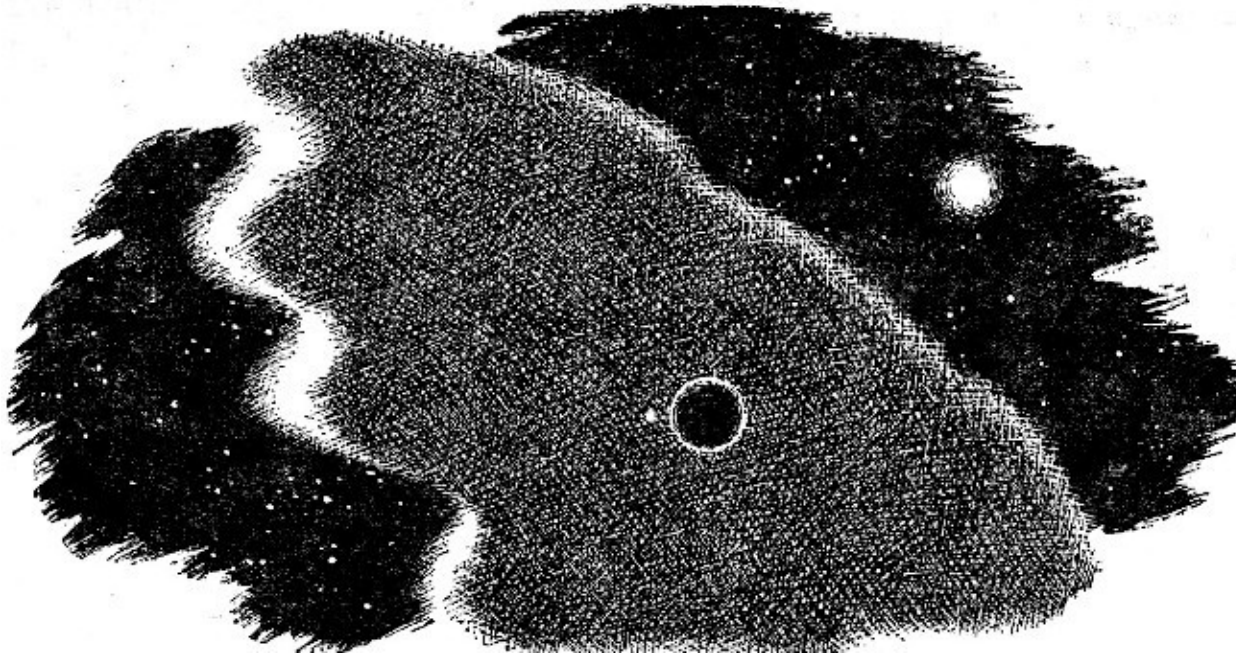
Riki maintained stoutly that she could feel the warmth from the sky, but that was improbable. But certainly heat did come from somewhere. The thermometer did not fall at all, that night. It rose. It was up to fifty below zero at dawn. During the day—they sent out twenty more bombs that second day—it was up to twenty degrees below zero. By the day after, there was highly competent computation from the home planet, and the concrete results of abstruse speculation, and the third day's bombs were placed with optimum spacing for heating purposes.

And by dawn of the fourth day the air was a balmy five degrees below zero, and the day after that there was a small running stream in the valley at midday.

There was talk of stocking the stream with fish, on the morning the Survey ship came in. The great landing-grid gave out a deep-toned, vibrant, humming note, like the deepest possible note of the biggest organ that could be imagined. A speck appeared very, very high up in a pale-blue sky with trimmings of golden gas-clouds. The Survey ship came down and down and settled as a shining silver object in the very center of the gigantic red-painted landing-grid.

Later, her skipper came to find Massy. He was in Herndon's office. The skipper struggled to keep sheer blankness out of his expression.

"What ... what the hell?" he demanded querulously of Massy. "This is the damnest sight in the whole galaxy, and they tell me you're responsible! There've been ringed planets before, and there've been comets and who-knows what! But shining gas pipes aimed at the sun, half a million miles across.... What the? There are two of them! Both the occupied planets!"



Herndon explained with a bland succinctness why the curtains hung in space. There was a drop in the solar constant—

The skipper exploded. He wanted facts! Details! Something to report! And dammit, he wanted to know!

Massy was automatically on the defensive when the skipper shot his questions to him. A Senior Colonial Survey officer is not revered by the Survey ship-service officers. Men like Massy can be a nuisance to a hard-working ship's officer. They have to be carried to unlikely places for their work of checking over colonial installations. They have to be put down on hard-to-get-at colonies, and they have to be called for, sometimes, at times and places which are inconvenient. So a man in Massy's position is likely to feel unpopular.

"I'd just finished the survey here," he said defensively, "when a cycle of sunspot cycles matured. All the sunspot periods got in phase, and the solar constant dropped. So I naturally offered what help I could to meet the situation."

The skipper regarded him incredulously.

"But ... it couldn't be done!" he said blankly. "They told me how you did it, but ... it couldn't be done! Do you realize that these vapor-curtains will make fifty borderline worlds fit for use? Half a pound of sodium-vapor a week!" He gestured helplessly. "They tell me the amount of heat reaching the surface here has been upped by fifteen per cent! D'you realize what *that* means?"

"I haven't been worrying about it," admitted Massy. "There was a local situation and something had to be done. I ... er ... remembered things, and Riki suggested something I mightn't have thought of, and it's worked out like this." Then he said abruptly: "I'm not leaving. I'll get you to take my resignation back. I ... I think I'm going to settle here. It'll be a long time before we get really temperate-climate conditions here, but we can warm up a valley like this for cultivation, and ... well ... it's going to be a rather satisfying job. It's a brand-new planet with a brand-new ecological system to be established——"

The skipper of the Survey ship sat down hard. Then the sliding door of Herndon's office opened and Riki came in. The skipper stood up again. Massy rather awkwardly made the introduction. Riki smiled.

"I'm telling him," said Massy, "that I'm resigning from the Service to settle down here."

Riki nodded. She put her hand in proprietary fashion on Massy's arm. The Survey skipper cleared his throat.

"I'm not going to take it," he said doggedly. "There've got to be detailed reports on how this business works. Dammit, if vapor-clouds in space can be used to keep a planet warm, they can be used to shade a planet, too! If you resign, somebody else will have to come out here to make observations and work out the details of the trick! Nobody could be gotten here in less than a year! You need to stay here to build up a report—and you ought to be available for consultation when this thing's to be done somewhere else! I'll report that I insisted as a Survey emergency——"

Riki said confidently:

"Oh, that's all right! He'll do that! Of course! Won't you?"

Massy nodded dumbly. He thought, *I've been lonely all my life. I've never belonged anywhere. But nobody could possibly belong anywhere as*

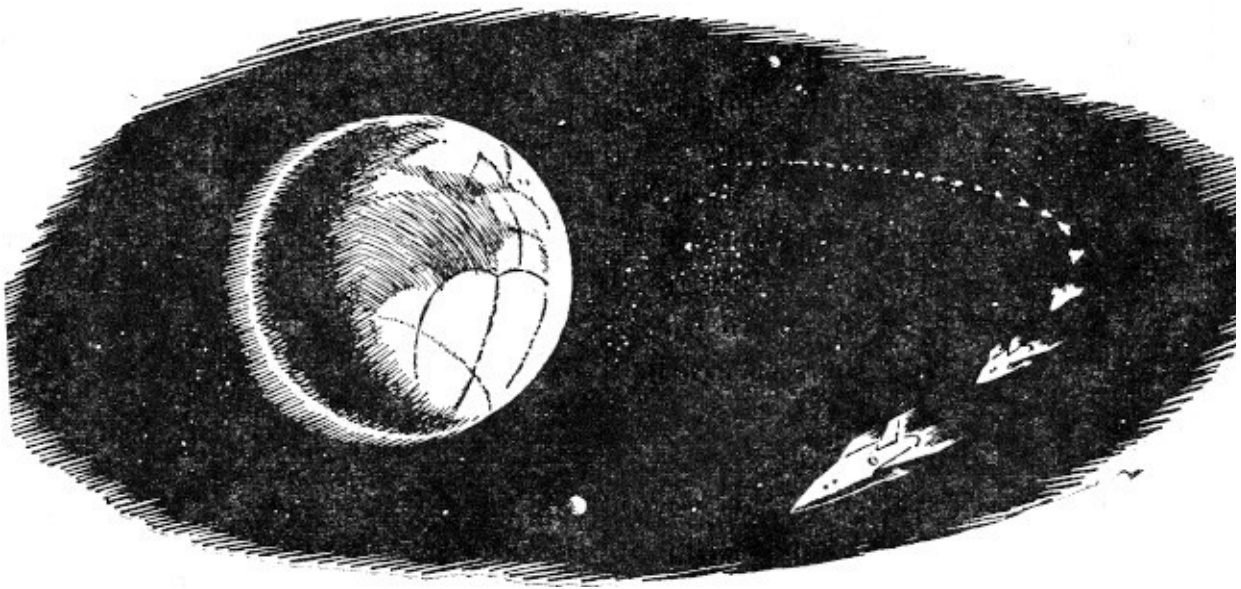
thoroughly as I'll belong here when it's warm and green and even the grass on the ground is partly my doing. But Riki'll like for me still to be in the Service. Women like to see their husbands wearing uniforms.

Aloud he said:

"Of course. It ... really needs to be done. Of course, you realize that there's nothing really remarkable about it. Everything I've done has been what I was taught, or read in books."

"Hush!" said Riki. "You're wonderful!"

THE END



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